Luke Sile

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U.S. Citizen (ITAR Compliant / Security Clearance Eligible)

Education

University of Michigan – Ann Arbor, College of Engineering Expected Dec. 2027
Bachelor of Science in Engineering in Aerospace Engineering, Minor in Electrical Engineering Ann Arbor, MI
- GPA: 3.5/4.0
- Honors: Sigma Gamma Tau (Aerospace Engineering Honor Society), Dean's Honors List
- Relevant Coursework: Electronic Circuits, Embedded Systems, Aerospace Structures, Dynamics & Vibrations, Gas Dynamics, Systems Engineering Principles, Computing Fundamentals

Engineering Experience

Intelligent Robotics and Autonomy Laboratory (iRaL) May 2026 – Present
Research Assistant Ann Arbor, MI
- Developed hardware for the laboratory's third-generation omnidirectional, over-actuated UAV designed to hover in any orientation, focusing on mechanical and electrical integration.
- Designed mechanical and structural components in SOLIDWORKS, assemble prototype hardware, and conduct subsystem integration and validation testing for autonomous flight experiments.
- Integrated, calibrated, and evaluated LiDAR and stereo camera perception systems supporting localization, mapping, and autonomous navigation research.
- Integrating a newly funded robotic arm payload onto the UAV alongside a Michigan State University research lab, owning hardware and software integration for a tree-shaking end effector for automated apple harvesting.

M-CLOUD – Michigan Collins Launchable Operational Unmanned Drone (X88) Sep. 2026 – Present
Responsible Engineer, Flight Computer (Controls) Ann Arbor, MI
- Own the flight computer for a Collins Aerospace (RTX) sponsored cooperative ISR system pairing an autonomous 5 kg fixed-wing search aircraft, COTS quadcopters, and a ground control station.
- Led a weighted trade study of 4 autopilots, screening against 6 go/no-go constraints (1 s manual RC override, 6+ PWM outputs, 0–40 °C) and scoring 5 pairwise-weighted criteria with weighted-sum and TOPSIS methods plus sensitivity checks.
- Selected a Pixhawk 6X on a Jetson Orin Nano baseboard (TOPSIS 0.829, ranked 1st) for triple redundant IMUs, an independent override processor, and an onboard companion computer at $0 marginal cost from team inventory.
- Decomposed Level 0 mission requirements into traceable subsystem requirements across autonomy, communications, and ground-station functions, including a 250 m line-of-sight link and full mission data logging.

Michigan Defense Tech (MDT) Jan. 2026 – Present
Operation Nightcrawl (Aug. 2026 – Present) & Project Lead, DARPA ASIMOV (Jan. – May 2026) Ann Arbor, MI
- Core member of a 13-person AG3-sponsored team building a drone that autonomously navigates, maps, and detects targets in dark, GPS-denied spaces, with a final integrated demo in April 2027.
- Led a six-member team on DARPA ASIMOV, co-authoring a System Requirements Document (SRD) translating Military Operational Values into traceable Developmental and Operational Test (DT/OT) requirements through System Requirements Review.
- Developed Python-based simulation environments and algorithmic testbeds incorporating Law of Armed Conflict (LOAC) and Rules of Engagement (ROE) logic to evaluate autonomous agent behavior.
- Member of the founding class of the Swarm Drone subteam, working on decentralized multi-agent UAV autonomy and a simulation-to-flight development pipeline.

BSA CubeSat Demonstration Program Jan. 2026 – May 2026
Avionics & Power Systems Engineer Ann Arbor, MI
- Designed, built, and flew a 1U CubeSat-format system on a tethered balloon, from requirements through flight and post-flight data analysis.
- Designed the 9 V to 5 V power system and a custom 2-layer Altium PCB around an RP2040 microcontroller and BMP388 sensor over I2C, verified by breadboard and continuity testing.
- Wrote and unit-tested C++ flight code logging sensor data to microSD at $\geq$1 Hz with error handling and status LED, verified against mass (<250 g), 1U form-factor, and 20-minute endurance requirements.

Leadership

Khajra Model Rocketry Apr. 2025 – Present
Co-Founder & Outreach Lead Ann Arbor, MI
- Co-founded a student high-power rocketry organization launching 4-ft rockets to ∼1,400-ft apogees; coordinate Tripoli Level 1 certification and launch operations.

Black Students in Aerospace (BSA) Mar. 2026 – Present
Conference Planning Chair Ann Arbor, MI
- Coordinated logistics, budgeting, and travel planning for 20+ students attending national aerospace conferences including AIAA SciTech, ASCEND, and NSBE.

Technical Skills

Hardware: PCB Design/Fabrication, Embedded Systems, Microcontroller Programming, Sensor Integration, LiDAR, Stereo Cameras, Oscilloscopes, Multimeters, Power Distribution, Soldering, Altium
Software & CAD: Python, MATLAB, C/C++, Arduino, PX4/ArduPilot, Linux, CATIA, SOLIDWORKS, Siemens NX, Star-CCM+, Git
Systems Engineering: Requirements Development, Verification & Validation (V&V), HITL/SITL Testing, DT/OT Frameworks, Trade Studies (Weighted Sum, TOPSIS), Interface Management, Root Cause Analysis
Certifications: INCOSE Associate Systems Engineering Professional (ASEP) – Anticipated May 2027
Languages: French (Native), Spanish (Proficient)